

StockMind: Financial Stock and Options Tracker

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1. Executive Summary and User Story

1.1. Introduction

Since the dawn of time in human history, people have always wanted to trade things and make money to provide for their families. In today's modern society, the stock market is the main place where this trade is. This document serves as the formal project proposal for our mobile application which we will be working on and building over the entire summer semester for this class. Over the last few years, especially after the pandemic happened, retail investing has become very popular with normal everyday people. But the big problem that is happening is that many beginners find it very hard to access good financial models. Supposably, the big Wall Street hedge funds have massive desktop computers and expensive software like Bloomberg terminals that cost thousands of dollars every single month just to use. Normal people do not have that kind of money laying around. So, we wanted to build Market Minds. Market Minds is our application because it will help them learn faster and make better choices without needing expensive desktop software's. We think that everyone should have the right to understand their financial future, regardless of if they went to a fancy finance college or not. By giving them a tool that can do heavy math, they will not just be guessing when they buy stocks, but they will know what the math says.

1.2. Example User Application

To fulfill the requirement of giving us an example of how the app will be used in a real-life situation, we have created a user story. Imagine a retail investor named John. John is thirty years old and he works a normal nine to five job in the city doing marketing. Every single morning, John wakes up at six o'clock in the morning, drinks his coffee, and he has to walk fifteen minutes to the train station. He takes the 7:45 AM express train to work. The train is always very crowded and there is no room to sit down most of the time, so he has to stand up holding the rail. Because he is commuting and standing on a moving train, his phones are way more accessible than his laptop computer. You cannot open a laptop while standing on a crowded train without dropping it. So, he pulls out his Android mobile phone. He opens the Market Minds application on his phone. The app will immediately show the livestock data which we get from API calls to Yahoo Finance. John wants to check on his Apple and Microsoft stocks because he heard on the news that technology is doing well today. He sees the line chart and it looks very sleek. Then, he thinks about maybe buying a call option to make some extra money. He navigates to the options section. He inputs a strike price to see if he should buy the call option, and the app runs a Binomial Model in the background to show the theoretical price on the screen. Because math runs very fast, John can make his decision before his train even arrives at his stop. This will definitely grab the user's attention because it provides real math on the go and shows exactly how a normal person would use this in their everyday daily routine.

2. Product Definition and Vision

2.1. The "Product Thinking" Framework

Our professor taught us about the product thinking framework, which is a very important way to design software before you even write any code. We are using the exact product thinking layout from the grading document to define our main goals for the whole summer semester:

- To make high level financial math and options calculators accessible to normal people (vision)

- our product will solve the everyday retail investors and beginners (target audience)
- problem of not having complex options models outside of their house or office (user problem)
- by giving them a mobile application strategy built on Android (strategy).
- We know if our product works when we see users running complex calculations fast without the app freezing or crashing their phone (goals).

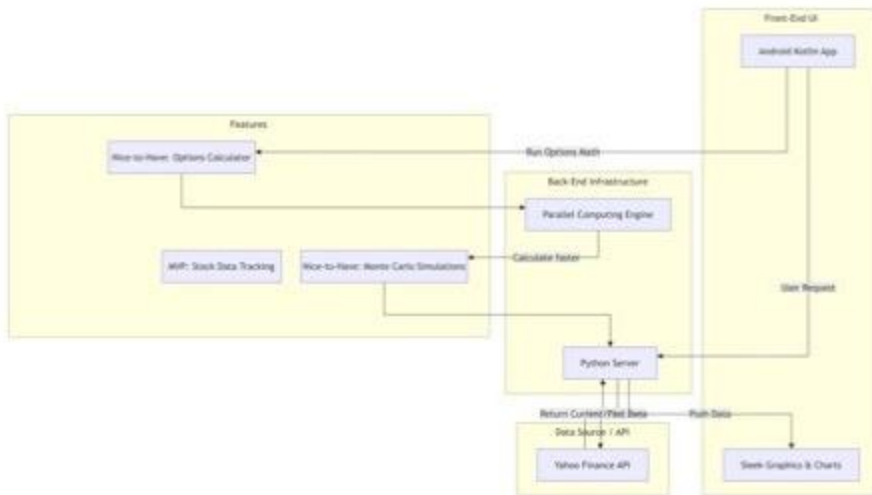
2.2. Unique Value Proposition

The unique value proposition is a business term that means what makes our app different from all the other apps on the Google Play store. There is a lot of stock apps out there like Robinhood or Webull, but they mostly just show you the price. Our unique value proposition is combining real-time market data with parallel computing options pricing directly on an Android phone. Parallel computing is when the computer processor does multiple math equations at the exact same time instead of doing them one by one in a line. This is as if you are cooking breakfast and you cook the eggs and the toast at the same time instead of waiting for the eggs to finish before putting the bread in the toaster. By doing this in the Python back end, we can do heavy Monte Carlo simulations that usually take a long time but do them very fast, so the mobile user doesn't have to wait.

2.3. Project Goals and Accountability

Setting these goals provides accountability for our team because we know the product only works if the math loads quickly. If we build the app and the user presses the button to calculate the Black Scholes model and the phone just shows a loading spinner for five minutes, then the user will just close the app and delete it. So, our main goal of accountability is making sure parallel computing actually reduces the time it takes to get the answer back to the phone screen.

3. System Architecture and Technologies



3.1. Mobile Application Strategy

When we were trying to figure out what platform to use, we chose a mobile app strategy because our target audience of users will often be outside. We want an app instead of a mobile-friendly website because we need native phone performance for the UI. If you use a website on a phone browser like Safari or Chrome, sometimes the buttons are too small, or the scrolling feels very weird and choppy. Plus, doing heavy math calculations is more complicated to do from a browser because the browser limits how much memory you can use. A native mobile app can connect to the server much smoother and provide a better experience.

3.2. Front-End User Interface

For the Front-End UI, we are going to use Android Kotlin. Kotlin is a programming language that is used to make Android apps and it is very good for making sleek graphics and buttons. Over the weekend, the team is going to draw up rough bad sketches of exactly what we are thinking for the layouts. We will draw where the buttons go, where the stock chart goes, and where the text boxes for the options calculator will be placed on the screen. Then we will write the Kotlin code to make those drawings turn into a real app.

3.3. Back-End Infrastructure

The Back-End infrastructure will be a Python server. Python is a programming language that is extremely popular for data science and complex mathematics. Because we must do things like the Binomial Model and Monte Carlo simulations, Python has a lot of libraries that make doing this math easier. The Python server will sit on the internet and wait for the Android phone to ask a question. When the phone asks to do a calculation, the Python server will use its parallel computing architecture to run the math on multiple cores and then send the final number back down to the phone.

3.4. Data Sources and APIs

For the Data API, we will make API calls to Yahoo Finance. An API stands for Application Programming Interface. It is basically like a waiter at a restaurant. When you go to a restaurant, you do not go into the kitchen and cook the food yourself. You tell the waiter what you want, the waiter goes to the kitchen and then brings the food back to your table. In our app, our Python server is the customer, the API is the waiter, and Yahoo Finance is the kitchen. We will ask the Yahoo Finance API for the current stock data and the past historical stock data, and it will bring us the JSON data so we can put it on our charts.

4. Project Plan and Timeline

4.1. Minimum Viable Product (MVP)

In software engineering, the MVP is the Minimum Viable Product. This means it is the smallest, most basic version of the app that actually works and does something useful. For our project timeline, the MVP is just getting the data pipelines working from the internet to the phone. This means we have to successfully make the API calls to Yahoo Finance, get the data back, parse the JSON, and displaying the live prices and past trends clearly to the user with basic code and graphics on the Kotlin front end. If we just get the stock prices to show up correctly on the phone screen without crashing, we will consider the MVP to be a complete success.

4.2. Post-MVP (Nice-to-Have Features)

If we have extra time after the MVP is finished, we will implement more things that are nice to have. First, we will add option trackers to see if the options are in the money, out the money, or knocked out. Second, we will add an options price calculator. This calculator will be the thing that actually does the Black Scholes math formula or the Binomial Model and the monte carlo simulations. The Black Scholes model is a very famous mathematical equation used in finance to find the price of European style options. The Monte Carlo simulation is named after a famous casino in Monaco because it uses randomness to predict different paths a stock could take in the future. Because running thousands of random paths takes a lot of computer power, we are planning to use the parallel computing we talked about earlier to reduce the calculation time for these simulations. These additions seem cool and hard but they are pretty simple to implement once the basic MVP data pipeline is set up.

5. Team Competencies and Work Distribution

To make sure the project gets done on time before the final presentation, we split the work distribution based on the specific strengths of each student in the team. By giving everyone a job they are already good at, we will not waste time trying to learn everything from scratch. We will all collaborate together independently on Wednesdays during the scheduled class period to make sure our different parts are connecting properly.

- JL : This team member has a strong background in backend development and system architecture. They will be responsible for building the Python backend server, defining the API endpoints, and handling communication between the mobile application and the server.
- Ishpreet : This team member has experience with user interface design and visual development. They will be responsible for creating the initial screen layouts and implementing the Android front end using Kotlin to provide a clean and user-friendly experience.

- ____WG____: This team member has strengths in algorithms, data structures, and data processing. They will be responsible for retrieving and processing data from Yahoo Finance, working with JSON responses, and implementing the mathematical models and simulations used by the application, such as option pricing calculations and Monte Carlo methods.

All of the work that we have detailed in this proposal document will serve as our strict roadmap for the summer semester. By following this plan, we will ensure that MarketMinds becomes a fully functioning prototype by the time the final presentation rolls around in August.